

- Description

This module manages a smart wind turbine blade angle adjustment system, ensuring that:

1. **The blade angle (pitch) is dynamically adjusted to optimize wind energy capture.**
2. **If wind speed exceeds the maximum safety threshold, the blades are adjusted to reduce stress and prevent damage.**
3. **If wind speed is too low, the blades are adjusted to maximize efficiency.**
4. **A log of all wind speed readings, blade adjustments, and efficiency metrics is maintained for performance monitoring.**

This system helps improve wind energy utilization, prevent turbine damage, and optimize power generation efficiency.

- Specification

module WindTurbineBladeAdjustment

var currentWindSpeed, maxWindSpeed, minWindSpeed, bladeAngle: real

var log: sequence of string

process updateWindSpeed(changeAmount: real)

functional scenario form:

currentWindSpeed ≤ maxWindSpeed && bladeAngle = 30.0, log = log ⊕ ("Wind Speed Updated: " + currentWindSpeed + " m/s, Blade at " + bladeAngle + "°"),
Wind speed updated, blade adjusted.

||

currentWindSpeed > maxWindSpeed && bladeAngle = 75.0, log = log ⊕ ("High Wind Warning: " + currentWindSpeed + " m/s, Adjusting Blade to " + bladeAngle + "°"),
High wind warning triggered, blade adjusted for safety.

||

currentWindSpeed < minWindSpeed && bladeAngle = 15.0, log = log ⊕ ("Low Wind Alert: " + currentWindSpeed + " m/s, Adjusting Blade to " + bladeAngle + "°"), Low wind alert triggered, blade adjusted for efficiency.

end process

end module

- Java Code

```
package code;
import java.util.*;
class WindTurbine42 {
    private double maxWindSpeed;
    private double minWindSpeed;
    private double currentWindSpeed;
    private double bladeAngle;
    private List<String> log;
    public WindTurbine42(double maxWindSpeed, double minWindSpeed, double initialWindSpeed) {
```

```

        this.maxWindSpeed = maxWindSpeed;
        this.minWindSpeed = minWindSpeed;
        this.currentWindSpeed = initialWindSpeed;
        this.bladeAngle = 30.0;
        this.log = new ArrayList<>();
    }

    public void updateWindSpeed(double changeAmount) {
        currentWindSpeed += changeAmount;
        adjustBladeAngle();
        String message = String.format("Wind Speed: %.2f m/s | Blade Angle: %.1f°", currentWindSpeed, bladeAngle);
        if (currentWindSpeed > maxWindSpeed) {
            message += " | WARNING: High Wind Speed! Adjusting Blades to Reduce Stress.";
        } else if (currentWindSpeed < minWindSpeed) {
            message += " | ALERT: Low Wind Speed! Adjusting Blades to Maximize Efficiency.";
        }
        System.out.println(message);
        log.add(message);
    }

    private void adjustBladeAngle() {
        if (currentWindSpeed > maxWindSpeed) {
            bladeAngle = 75.0;
        } else if (currentWindSpeed < minWindSpeed) {
            bladeAngle = 15.0;
        } else {
            bladeAngle = 30.0;
        }
    }

    public void printLog() {
        System.out.println("\nWind Turbine Adjustment Log:");
        for (String logEntry : log) {
            System.out.println(logEntry);
        }
    }
}

public class c42_WindEnergySystem {
    public static void main(String[] args) {
        WindTurbine42 turbine = new WindTurbine42(25.0, 5.0, 12.0);
        // testcase-VT:
        // turbine.updateWindSpeed(10.0);
        // turbine.updateWindSpeed(-8.0);
        // turbine.updateWindSpeed(-10.0);
    }
}

```

```
//      turbine.updateWindSpeed(15.0);
//      turbine.updateWindSpeed(10.0);

//      // testcase-FT:
//      turbine.updateWindSpeed(-1.7976931348623157E308);
//      turbine.updateWindSpeed(1.1209145429141497E308);
//      turbine.updateWindSpeed(1.7976828080405392E308);
//      turbine.updateWindSpeed(0.0);
//      turbine.updateWindSpeed(-6.838283689477282E307);

//      // testcase-Z3:
//      turbine.updateWindSpeed(10);
//      turbine.updateWindSpeed(8);
//      turbine.updateWindSpeed(10);
//      turbine.updateWindSpeed(15);
//      turbine.updateWindSpeed(10);

//      // testcase-UVT:
turbine.updateWindSpeed(10.0);
turbine.updateWindSpeed(-8.0);
turbine.updateWindSpeed(-10.0);
turbine.updateWindSpeed(15.0);
turbine.updateWindSpeed(10.0);

//      turbine.printLog();
    }
}
```

- Coverage report

1. Test case number: 5
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.2 %	10	832	842
▼ src	1.2 %	10	832	842
▼ code	1.2 %	10	832	842
▼ c42_WindEnergySystem.java	100.0 %	10	0	10
> c42_WindEnergySystem		0	0	0
> WindTurbine42	100.0 %	10	0	10

3. Vibration testing instruction coverage: 97.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.7 %	141	20,586	20,727
▼ src	0.7 %	141	20,586	20,727
▼ code	0.7 %	141	20,586	20,727
▼ c42_WindEnergySystem.java	97.9 %	141	3	144
> c42_WindEnergySystem	89.3 %	25	3	28
> WindTurbine42	100.0 %	116	0	116

4. Fuzz testing branch coverage: 80%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	1.0 %	8	834	842
✓ src	1.0 %	8	834	842
✓ code	1.0 %	8	834	842
✓ c42_WindEnergySystem.java	80.0 %	8	2	10
> WindTurbine42	80.0 %	8	2	10
> c42_WindEnergySystem		0	0	0

5. Fuzz testing instruction coverage: 95.8%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.7 %	138	20,589	20,727
✓ src	0.7 %	138	20,589	20,727
✓ code	0.7 %	138	20,589	20,727
✓ c42_WindEnergySystem.java	95.8 %	138	6	144
> c42_WindEnergySystem	89.3 %	25	3	28
> WindTurbine42	97.4 %	113	3	116

6. Z3 branch coverage: 60%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.7 %	6	834	840
✓ src	0.7 %	6	834	840
✓ code	0.7 %	6	834	840
✓ c42_WindEnergySystem.java	60.0 %	6	4	10
> WindTurbine42	60.0 %	6	4	10
> c42_WindEnergySystem		0	0	0

7. Z3 instruction coverage: 79.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.5 %	113	22,265	22,378
✓ src	0.5 %	113	22,265	22,378
✓ code	0.5 %	113	22,265	22,378
✓ c42_WindEnergySystem.java	79.6 %	113	29	142
> WindTurbine42	77.6 %	90	26	116
> c42_WindEnergySystem	88.5 %	23	3	26

8. Unimproved V-method branch coverage: 80%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	1.0 %	8	832	840
✓ src	1.0 %	8	832	840
✓ code	1.0 %	8	832	840
✓ c42_WindEnergySystem.java	80.0 %	8	2	10
> WindTurbine42	80.0 %	8	2	10
> c42_WindEnergySystem		0	0	0

9. Unimproved V-method instruction coverage: 84.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.5 %	120	22,258	22,378
✓ src	0.5 %	120	22,258	22,378
✓ code	0.5 %	120	22,258	22,378
✓ c42_WindEnergySystem.java	84.5 %	120	22	142
> WindTurbine42	83.6 %	97	19	116
> c42_WindEnergySystem	88.5 %	23	3	26